



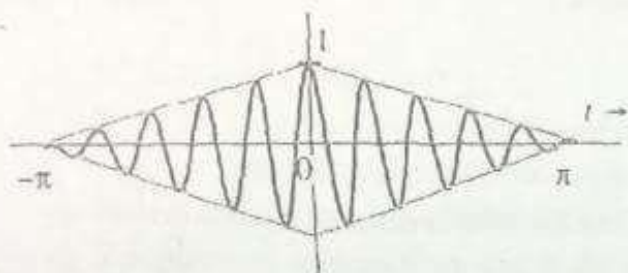
Course:	Analog and Digital Communication	Course Code:	EL323
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Instruction/Notes:

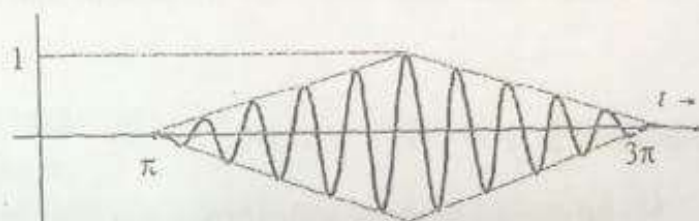
Question # 1 [9+6] CLO1

The signals in the following figure are modulated signals with carrier $\cos(10t)$. $2\pi f = \omega$

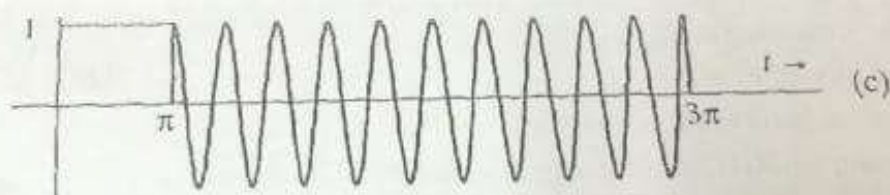
- Find the Fourier transform of these signals using the appropriate properties of the Fourier transform (use the table attached at the last page).
- Sketch the amplitude and phase spectra for figure (a) and (b)



(a)



(b)



(c)

Question # 2 [10 + 10] CLO2

- For the signal $g(x) = \text{Sinc}(k \cdot x)$, find the energy of the signal using Parseval's theorem.
- Generalize the Parseval's theorem to show that for real, Fourier transformable signals $g_1(t)$ and $g_2(t)$:

$$\int_{-\infty}^{\infty} g_1(t)g_2(t) dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} G_1(-\omega)G_2(\omega) d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} G_1(\omega)G_2(-\omega) d\omega$$

Question # 3 [4+4+3+4] CLO3

$\frac{1}{100}$ Am 10000t

For each of the following baseband signals:

(i) $m(t) = 2\cos(1000t)$ (ii) $m(t) = 4\cos 1000t + 2\cos 2000t$

(iii) $m(t) = 2\cos(1000t)\cos(3000t)$

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- Sketch the spectrum of $m(t)$
- Sketch the spectrum of DSB-SC signal $m(t) \cdot \cos(10,000t)$
- Identify USB and LSB spectra
- Identify the frequencies in the baseband and the corresponding frequencies in the DSB-SC, USB and LSB spectra. Explain the nature of frequency shifting in each case.

Question # 4 [5] CLO4

Determine the power efficiency (η) and the percentage of the total power carried by the sidebands of the AM wave for tone modulation when modulation index is 0.5 and 0.3.

$\eta = \frac{P_{sidebands}}{P_{total}} = \frac{2\mu^2}{1+\mu^2}$

Question # 5 [10+10] CLO5

- a) An angle-modulated signal with carrier frequency $\omega_c = 2\pi \times 1000000$ is described by:

$\phi_{EM}(t) = 10\cos(\omega_c t + 0.1\sin(1000)\pi t)$

- Find the power of the modulated signal. 3 (36)
- Find the frequency deviation Δf .
- Find the phase deviation $\Delta\phi$.
- Estimate the bandwidth of $\phi_{EM}(t)$.
- Considering $\phi_{EM}(t)$ as a PM signal with $k_p = 10$, find $m(t)$.
- Considering $\phi_{EM}(t)$ as an FM signal with $k_f = 10\pi$, find $m(t)$.

$10 \int m(t) dt = 0.1 \sin 1000\pi t$
 $\cos 1000\pi t$
 $10 \times \frac{1000\pi}{1000\pi} = 10$

- b) Design an Armstrong indirect FM modulator to generate an FM carrier with a carrier frequency of 96 MHz and $\Delta f = 20$ kHz. A narrow-band FM generator with $f_c = 200$ kHz and adjustable Δf in the range of 9 to 10 Hz is available. The stock room also has an oscillator with adjustable frequency in the range of 9 to 10 MHz. There is a bandpass filter with any required center frequency, and only frequency doublers are available.

Handwritten calculations for part b):
 $64 \times 32 = 2048$
 $9.765 \text{ MHz} \times 2 = 19.53 \text{ MHz}$
 $12.8 \text{ MHz} \times 3 = 38.4 \text{ MHz}$
 $16 \times 32 = 512$
 $32 \times 16 = 512$
 $100, 200, 400, 800, 1600, 3200, 6400, 12800, 25600, 51200, 102400, 204800, 409600, 819200, 1638400, 3276800, 6553600, 13107200, 26214400, 52428800, 104857600, 209715200, 419430400, 838860800, 1677721600, 3355443200, 6710886400, 13421772800, 26843545600, 53687091200, 107374182400, 214748364800, 429496729600, 858993459200, 1717986918400, 3435973836800, 6871947673600, 13743895347200, 27487790694400, 54975581388800, 109951162777600, 219902325555200, 439804651110400, 879609302220800, 1759218604441600, 3518437208883200, 7036874417766400, 14073748835532800, 28147497671065600, 56294995342131200, 112589990684262400, 225179981368524800, 450359962737049600, 900719925474099200, 1801439850948198400, 3602879701896396800, 7205759403792793600, 14411518807585587200, 28823037615171174400, 57646075230342348800, 115292150460684697600, 230584300921369395200, 461168601842738790400, 922337203685477580800, 1844674407370955161600, 3689348814741910323200, 7378697629483820646400, 14757395258967641292800, 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1447401115466452442794637312608598848165874808320507050493219800098914120499200, 2$

6 [4+6] CLO6

and prove the Sampling theorem.

A signal $g(t) = \text{sinc}^2(5\pi t)$ is sampled (using uniformly spaced impulses) at a rate of (i) 5 Hz (ii) 10 Hz (iii) 20 Hz. For each of the three case:

- 1) Sketch the sampled signal
- 2) Sketch the spectrum of the sampled signal
- 3) Explain whether you can recover the signal $g(t)$ from the sampled signal.
- 4) If the sampled signal is passed through an ideal low-pass filter of bandwidth 5 Hz, sketch the spectrum of the output signal.

Question # 7 [3+4+4+4] CLO7

A compact disc (CD) records audio signals digitally by using PCM. Assume the audio signal bandwidth to be 15 kHz.

- a) What is the Nyquist rate?
- b) If the Nyquist samples are quantized into $L=65,536$ levels and then binary coded, determine the number of binary digits required to encode a sample.
- c) Determine the number of binary digits per second (bits/s) required to encode the audio signal.
- d) For practical reasons discussed in the text, signals are sampled at a rate well above the Nyquist rate. Practical CDs use 44,100 samples per second. If $L=65,536$, determine the number of bits per second required to encode the signal, and the minimum bandwidth required to transmit the encoded signal.

128

125

200

90m

164

100

0.96m

98

95